

Generation Interconnection System Impact Study Report

For Queue Project Q23-2356

Oceanside 500 kV

October 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2356. The Interconnection Customer proposes a 596.9 MW natural gas combined cycle facility interconnecting at the Oceanside 500 kV substation in the Michigan area. The request was received on November 04, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 601 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$57,500,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bg Oaks - Laureles 100.0 kV Line	Dekalb - Joes Chapel 500.0 kV	116.5	396	5.7
Sidney A Murray Jr Hydroelectric - 24Th & Gardendale Substation 230.0 kV Line	Richland - Dekalb 500.0 kV	116.1	1,060	8.6
Ward Gulf Substation - Laureles 100.0 kV	Jons Chapel - Dekalb 500.0 kV	111.4	439	5.7
Richland 500/230kV Transformer Bank	Troy Energy Llc - Richland 500.0 kV	108.9	3,175	29.8
Richland - Dekalb 500.0 kV	Breed - Sidney A Murray Jr Hydroelectric 230.0 kV	108.3	5,511	50.2
Minnehaha - Napoleon Peaking Sta 230.0 kV	Troy Energy Llc - Richland 500.0 kV	107.4	1,575	11.9
Bluebonnet - Jones Chapel 230.0 kV	Robson Park - Russell Fork 230.0 kV	105.3	2,175	42.5
Dekalb 500/1 kV Autotransformer	Dekalb - Jones Chael 500.0 kV	105.3	3,072	25.9
Dekalb 230/1kV Transformer Bank	Dekalb - Jones Chapel 500.0 kV	105.3	3,072	25.9
Sidney A Murray Jr Hydroelectric - Breed 230.0 kV	Richlnd - Dekalb 500.0 kV	103.4	2,969	24.1
Maple Street - Leetonia 230.0 kV	El Paso - Ravenna 500.0 kV	103.2	990	5.0
Atascocita - Maple St 230.0 kV	El Paso - Ravenna 500.0 kV	103.2	1,011	5.0
Big Oaks - Cypress Crk 100.0 kV	Dekalb - Jones Chapel 500.0 kV	103.0	422	5.7
Try Energy Llc - Richland 500.0 kV	Trego - Swan Vly Substation 500.0 kV L	102.6	5,380	35.1
Landsman Crk - 24Th & Gardendale 230.0 kV line	Ricland - Dekalb 500.0 kV	102.3	1,176	8.6
Omni-Hackberry - Norwich 500.0 kV	Trego - Beaver 500.0 kV	101.9	4,479	18.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Barrett - Wolfe 500.0 kV L	Beaver - Trego 500.0 kV L	101.5	3,714	9.2
Minnehaha - Station S Van Nuys 230.0 kV	Troy Energy Llc - Richland 500.0 kV	101.4	2,237	15.9
Pueblo Xtra 500 kV/230 kV Transformer	Pueblo Xtra - Boca Raton 500.0 kV	101.3	2,553	22.5
Sa S Van Nuys - Gould 230.0 kV	Troy Energy Llc - Richland Substation 500.0 kV	101.2	986	6.2
Muskingum River - Tidd Substation 500.0 kV Line	El Paso - Ravenna 500.0 kV	101.1	3,605	12.0
Napoleon Peaking Station Substation - Richland 230.0 kV	Richland - Troy Enrgy Llc 500.0 kV	100.8	1,643	12.0
Dekalb Substation - Jones Chapel 500.0 kV	Dekalb - Robison Park 230.0 kV	100.5	5,380	49.5
Joso - Onancock Diesel Generation Facility 230.0 kV	Beaver - Eufaula 500.0 kV	100.1	2,698	13.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bg Oaks - Laureles 100.0 kV Line	Rebuild line with higher-capacity conductor	\$3,104,000	13% (\$416,000)
Sidney A Murray Jr Hydroelectric - 24Th & Gardendale Substation 230.0 kV Line	Rebuild line with higher-capacity conductor	\$19,469,000	8% (\$1,663,000)
Laureles - Ward Gulf 100.0 kV	Rebuild line with higher-capacity conductor	\$6,038,000	13% (\$808,000)
Richland 500/230 kV XFMR	Replace transformer with higher-rated unit	\$23,554,000	9% (\$2,185,000)
Richland - Dekalb 500.0 kV Line	Reconductor line	\$49,878,000	9% (\$4,335,000)
Minnehaha - Napoleon Peaking Station 230.0 kV	Reconductor line	\$18,078,000	9% (\$1,655,000)
Bluebonnet - Jones Chapel 230.0 kV	Reconductor line	\$13,561,000	11% (\$1,464,000)
Dekalb 500/1 kV Autotransformer	Replace transformer with higher-rated unit	\$33,342,000	8% (\$2,521,000)
Dekalb 230/1kV Transformer Bank	Replace transformer with higher-rated unit	\$9,863,000	8% (\$746,000)
Sidney A Murray Jr Hydroelectric - Breed 230.0 kV	Reconductor line	\$20,265,000	8% (\$1,614,000)
Maple Street - Leetonia 230.0 kV	Reconductor line	\$4,114,000	12% (\$486,000)
Atascocita - Maple St 230.0 kV	Reconductor line	\$15,703,000	12% (\$1,854,000)
Big Oaks - Cypress Crk 100.0 kV Line	Reconductor line	\$2,403,000	13% (\$322,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Troy Energy Llc - Richland 500.0 kV	Reconductor line	\$84,273,000	8% (\$6,933,000)
Landsman Crk - 24Th & Gardendale 230.0 kV line	Reconductor line	\$21,514,000	8% (\$1,838,000)
Omni-Hackberry - Norwich 500.0 kV	Reconductor line	\$30,210,000	6% (\$1,930,000)
Barrett - Wolfe 500.0 kV line	Reconductor line	\$100,648,000	6% (\$6,200,000)
Minnehaha - Station S Van Nuys 230.0 kV	Reconductor line	\$3,495,000	8% (\$264,000)
Pueblo Xtra 500 kV/230 kV Auto	Replace transformer with higher-rated unit	\$54,178,000	14% (\$7,523,000)
Sa S Van Nuys - Gould 230.0 kV	Reconductor line	\$5,461,000	11% (\$593,000)
Muskingum River - Tidd Substation 500.0 kV Line	Reconductor line	\$92,216,000	7% (\$6,068,000)
Richland - Napoleon Peaking Sta 230.0 kV	Reconductor line	\$6,175,000	9% (\$565,000)
Dekalb - Jones Chapel 500.0 kV L	Reconductor line	\$49,447,000	10% (\$4,987,000)
Joso - Onancock Diesel Generation Facility 230.0 kV	Reconductor line	\$7,814,000	7% (\$530,000)
	Total allocated to project		\$57,500,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.